Salonee Verma

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EDUCATION

University of Virginia, Charlottesville, VA *August 2023 — May 2027*

B.A. in Computer Science, Concentration in AI & Cognitive Science, Concentration in Linguistics *GPA 3.96*

Awards & Honors: Jefferson Scholarship (merit-based full-ride for UVA), Echols Scholar (honors college), College Science Scholar

Relevant Coursework: Machine Learning, Artificial Intelligence, Natural Language Processing, Data Structures & Algorithms, Discrete Mathematics, Computational Theory, Computer Architecture, Operating Systems, Linear Algebra, Software Development Essentials

Organizations: HooHacks (President), Long Term Memory Lab, Human AI Technology Lab, Student Technology Council.

EXPERIENCE

Howard Hughes Medical Institute, Janelia Research Campus | Flybody, Dr. Srini Turaga *May 2026 — Present*

Machine Learning Research Intern Ashburn, VA

- Transformed fruitfly connectomic data to classify femoral chordotonal organs as flexion or extension based on connectivity.
- Developed preliminary inverse kinematics for implementing antennal grooming behaviors in a biomechanical fly model by virtualizing neural activity-based real fly grooming experiments and training the model using reinforcement learning.
- Migrated to NVIDIA Warp to cut training time by 50%; implemented realistic physical flight in the biomechanical fly model.
- Co-author on upcoming Turaga Lab & Google DeepMind paper in *Nature* due to contributions on modeling flight and mapping FeCOs.

Human AI Technology Lab | Robust Human-AI Interactions, Dr. Afsaneh Doryab *August 2025 — Present*

Undergraduate Research Assistant Charlottesville, VA

- Conducts research in basing human-AI interactions off of accurate human behavior models in trust & decision-making.
- Develops models of trust and collaboration within AI systems and contributes to experimentally proving their efficacy.

Long Term Memory Lab | Scalp Electroencephalographic Activity, Dr. Nicole Long *May 2025 — Present*

Undergraduate Research Assistant Charlottesville, VA

- Conducts EEG recording sessions, including participant preparation, electrode placement, and real-time signal monitoring.
- Supports research team with neuroimaging session coordination, data collection, and adherence to experimental protocols.

Chandra Robot Autonomy Lab | Control Barrier Functions, Dr. Rohan Chandra *February 2025 — May 2025*

Undergraduate Research Assistant Charlottesville, VA

- Executed and analyzed existing code implementations of NeRF-CBF and Depth-CBF with point clouds in order to benchmark and evaluate a new vision-based control barrier function, while collaborating with a team of undergraduate researchers.

Howard Hughes Medical Institute, Janelia Research Campus | FlyRewards, Dr. Srini Turaga *December 2023 — August 2024*

Machine Learning Research Intern Ashburn, VA

- Applied Eureka, a pipeline for generating reward functions for RL tasks using large language models, to the flybody model, a proprietary MuJoCo physics model and RL environment developed in collaboration with Turaga Lab and Google DeepMind.
- Evaluated current pipelines from Google DeepMind and NVIDIA for RL tasks concerning a *Drosophila melanogaster* model.

Howard Hughes Medical Institute, Janelia Research Campus | CortexConnoisseur, Dr. Srini Turaga *June 2023 — August 2023*

Machine Learning Research Intern Ashburn, VA

- Trained LLaMA 2 with PyTorch and LoRA on a custom cleaned dataset containing scraped data from academic publications.
- Developed CortexConnoisseur, a large language model capable of answering questions about computational neuroscience with minimal hallucination and better fine-tuned accuracy, through data mining and cleaning of 19 million academic papers.

PUBLICATIONS

First Author: Toward Human-Inspired Collaborative Intelligence for Human-AI Team Decision-Making: A Survey

- Bethapuri, Sree Deeksha and Cardei, Maria and Sprague, Hannah and **Verma, Salonee** and Doryab, Afsaneh, Toward Human-Inspired Collaborative Intelligence for Human-AI Team Decision-Making: A Survey (May 12, 2026). Available at SSRN: https://ssrn.com/abstract=6754660 or http://dx.doi.org/10.2139/ssrn.6754660

OTHER PROJECTS

Benchmarking LLMs on Abstracted Grammatical Understanding With Low-Resource Languages

- Developing a benchmark to determine if LLMs have abstract grammatical reasoning capabilities in non-standard grammars using low-resource languages, and determining whether LLMs are a viable solution for endangered language revitalization.
- Undergraduate Honors Thesis (Distinguished Major): supervised by Professor Yangfeng Ji, projected end date of May 2027.

Transformer-based Machine Translation for Speech Affected by Broca's Aphasia

- Developed a Transformer-based error correction pipeline to effectively translate raw inputs of speech affected by Broca's aphasia into standard English by fine-tuning Google's T5 with a grammar error correction model and data from AphasiaBank.
- Won 1st place at Robert H. Herndon Memorial Science Competition sponsored by the Aerospace Corporation.

SKILLS

Languages: C, C++, Python, Java, JavaScript, ReactNative, HTML/CSS, Assembly, MATLAB

Tools: Linux, Git, Docker, Bash, CMake, CUDA, Gym, IsaacGym, MuJoCo, LoRA

Libraries: JAX, NVIDIA Warp, PyTorch, Transformers, Keras, TensorFlow, OpenCV, React, Pandas, NumPy, Matplotlib, Datasets